

Pre-Algebra Practice Test IV - Version One

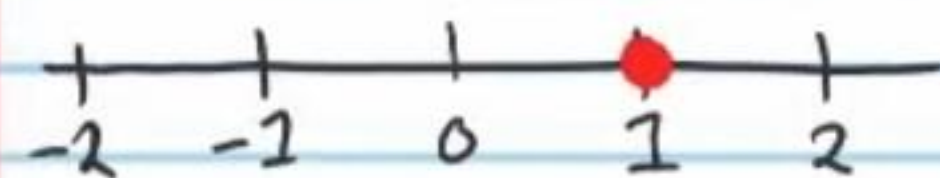
Name
Date
Course

Instructions

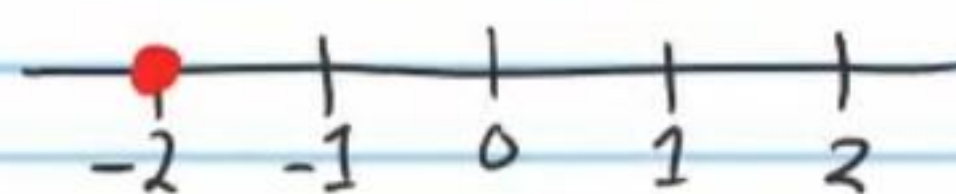
- As always, write a title, name, date, and course.
- Number each problem, circle the number, and show all work.
- Draw a rectangle around your answers.
- Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

Graph the following equations.

① a) $x = 1$



b) $x = -2$



Solve.

② $2.3x - 5.6x + 1 = 8.7x + 6$

$$-3.3x + 1 = 8.7x + 6$$

$$3.3x - 3.3x + 1 = 8.7x + 6 + 3.3x$$

$$1 = 12x + 6$$

$$-6 + 1 = 12x + 6 - 6$$

$$-5 = 12x$$

$$\frac{-5}{12} = \frac{12x}{12}$$

$$\boxed{-\frac{5}{12} = x}$$

- ③ If 49 is increased to 51, what is the percent increase? $\star$ Round to the nearest hundredth of a percent.

$$N = F + Fr$$

$$51 = 49 + 49r$$

$$-49 + 51 = 49 + 49r - 49$$

$$2 = 49r$$

$$\frac{2}{49} = \frac{49r}{49}$$

$$0.040816 \approx r$$

$$4.0816\% \approx \boxed{4.08\%}$$

Solve. Round to the nearest thousandth.

④ $-2.8(3z + 2) + 3(z - 4) = 3.1z - 2z$

$$-8.4z - 5.6 + 3z - 12 = 3.1z - 2z$$

$$-5.4z - 17.6 = 1.1z$$

$$-17.6 = 6.5z$$

$$-2.70769 \approx z$$

$$\boxed{-2.708 \approx z}$$

Find the algebraic solutions and the graphical solutions of the following inequalities.

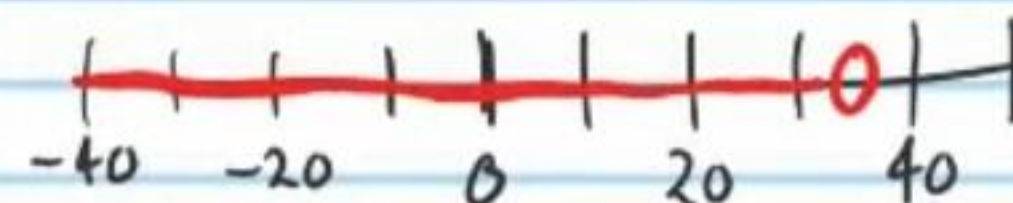
⑤ $\frac{x}{5} - 1 < 6$

$$1 + \frac{x}{5} - 1 < 6 + 1$$

$$\frac{x}{5} < 7$$

$$(5)\frac{x}{5} < 7(5)$$

$$\boxed{x < 35}$$



⑥ $-5 \geq 1 + 3y + 4y$

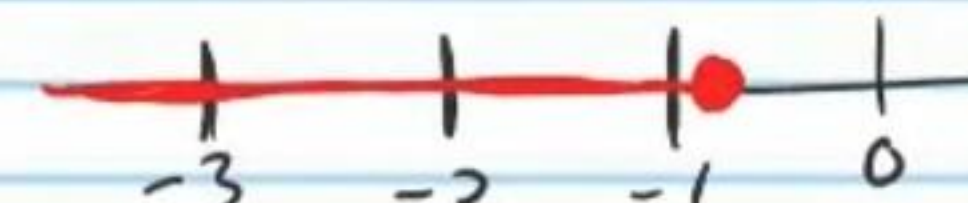
$$-5 \geq 1 + 7y$$

$$-1 - 5 \geq 1 + 7y - 1$$

$$-6 \geq 7y$$

$$\frac{-6}{7} \geq \frac{7y}{7}$$

$$\boxed{-\frac{6}{7} \geq y}$$



Solve.

$$\textcircled{7} -2z < 5z + 3 - 3z$$

$$-2z < 2z + 3$$

$$-2z - 2z < 2z + 3 - 2z$$

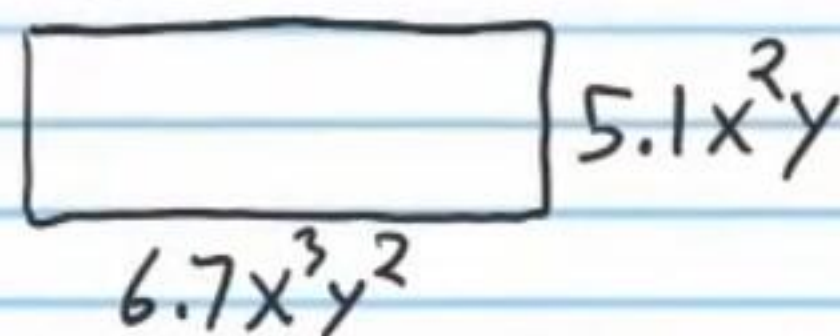
$$-4z < 3$$

$$\frac{-4z}{-4} > \frac{3}{-4}$$

$$z > -\frac{3}{4}$$



$\textcircled{9}$ Find the area of the rectangle below.



$$A = LW$$

$$A = (5.1x^2y)(6.7x^3y^2)$$

$$A = \boxed{34.17x^5y^3}$$

$\textcircled{8}$ What percent of 603 is 900? $\star$ Round to the nearest tenth of a percent.

$$603x = 900$$

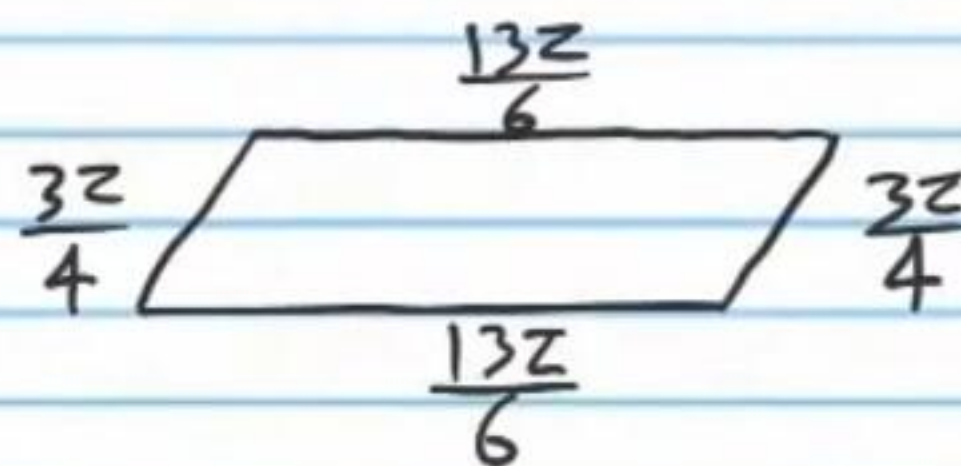
$$\frac{603x}{603} = \frac{900}{603}$$

$$x \approx 1.492537$$

$$149.2537\%$$

$$\boxed{149.3\%}$$

$\textcircled{10}$ Find the perimeter of the parallelogram below.



$$P = \frac{13z}{6} + \frac{13z}{6} + \frac{3z}{4} + \frac{3z}{4}$$

$$P = \frac{26z}{6} + \frac{6z}{4}$$

$$P = \frac{13z}{3} + \frac{3z}{2}$$

$$P = \frac{26z}{6} + \frac{9z}{6}$$

$$\boxed{P = \frac{35z}{6}}$$

$\textcircled{11}$ Find the surface area of a sphere with radius $\frac{3x^2y^3}{5z}$ (keep π as a factor).

$$A = 4\pi r^2$$

$$A = 4\pi \left(\frac{3x^2y^3}{5z}\right)^2$$

$$A = 4\pi \left(\frac{3x^2y^3}{5z}\right) \left(\frac{3x^2y^3}{5z}\right)$$

$$A = 4\pi \left(\frac{4x^4y^6}{25z^2}\right)$$

$$A = \frac{16\pi x^4y^6}{25z^2}$$

$\textcircled{13}$ a) Evaluate your answer for problem #11 if $y=1$, $x=2$, and $z=3$.

$$A = \frac{16\pi(2)^4(1)^6}{25(3)^2}$$

$$A = \frac{16\pi(16)(1)}{25(9)}$$

$$\boxed{A = \frac{256\pi}{225}}$$

$$\begin{array}{r} 16 \\ \times 16 \\ \hline 196 \\ 160 \\ \hline 256 \end{array}$$

$\textcircled{12}$ Find the average of the following numbers: $5x, -2x, 4x, 3x, 2x, 5x, -4x, -x$

$$\text{Mean} = \frac{5x - 2x + 4x + 3x + 2x + 5x - 4x - x}{8}$$

$$= \frac{12x}{8}$$

$$= \boxed{\frac{3x}{2}}$$

b) Find the volume of a sphere with radius 1 inch (keep π as a factor in your answer).

$$V = \frac{4\pi r^3}{3}$$

$$V = \frac{4\pi(1)^3}{3}$$

$$V = \frac{4\pi(1)}{3}$$

$$\boxed{V = \frac{4\pi}{3} \text{ in}^3}$$

- ⑭ The sum of four consecutive odd integers is 168. Identify these integers.

$$\begin{aligned} 1^{\text{st}} &= x \\ 2^{\text{nd}} &= x+2 \\ 3^{\text{rd}} &= x+4 \\ 4^{\text{th}} &= x+6 \end{aligned}$$

$$x + (x+2) + (x+4) + (x+6) = 168$$

$$4x + 12 = 168$$

$$-12 + 4x + 12 = 168 - 12$$

$$4x = 156$$

$$\boxed{39, 41, 43, 45}$$

$$\frac{4x}{4} = \frac{156}{4}$$

$$x = 39$$

- ⑮ If 27% of a number is 0.675, what is the number?

$$0.27x = 0.675$$

$$\frac{0.27x}{0.27} = \frac{0.675}{0.27}$$

$$\boxed{x = 2.5}$$

- ⑯ The size of Casey's hamburgers decreased by 21%. The new volume is 2.7 ounces. What was the volume before the decrease?
★ Round to the nearest tenth.

$$N = F - Fr$$

$$2.7 = F - F(0.21)$$

$$2.7 = F - 0.21F$$

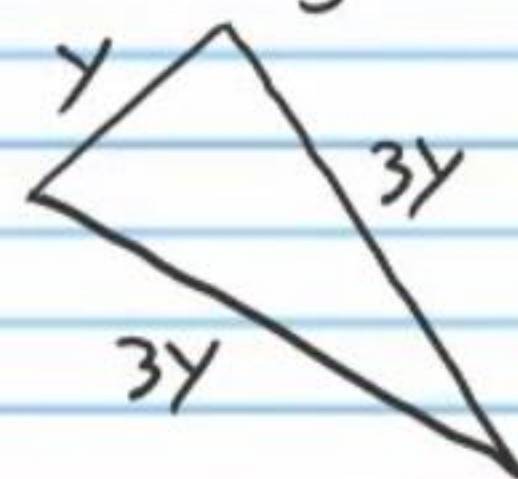
$$2.7 = 0.79F$$

$$\frac{2.7}{0.79} = \frac{0.79F}{0.79}$$

$$3.4177 \approx F$$

$$\boxed{3.4 \text{ oz} \approx F}$$

- ⑰ Find the perimeter of the triangle below.



$$P = y + 3y + 3y$$

$$\boxed{P = 7y}$$

- ⑱ You deposit money in a new savings account. The account pays 0.8% simple interest annually. After five years, the account grows to \$31.20. How much money did you deposit (assume no additional deposits)?

$$A = P + Prt$$

$$31.2 = P + P(0.008)(5)$$

$$31.2 = P + 0.04P$$

$$31.2 = 1.04P$$

$$\frac{31.2}{1.04} = \frac{1.04P}{1.04}$$

$$\boxed{\$30 = P}$$

$$\textcircled{19} -4y + 5 + 0.8y = -1.8y$$

$$-3.2y + 5 = -1.8y$$

$$3.2y - 3.2y + 5 = -1.8y + 3.2y$$

$$5 = 1.4y$$

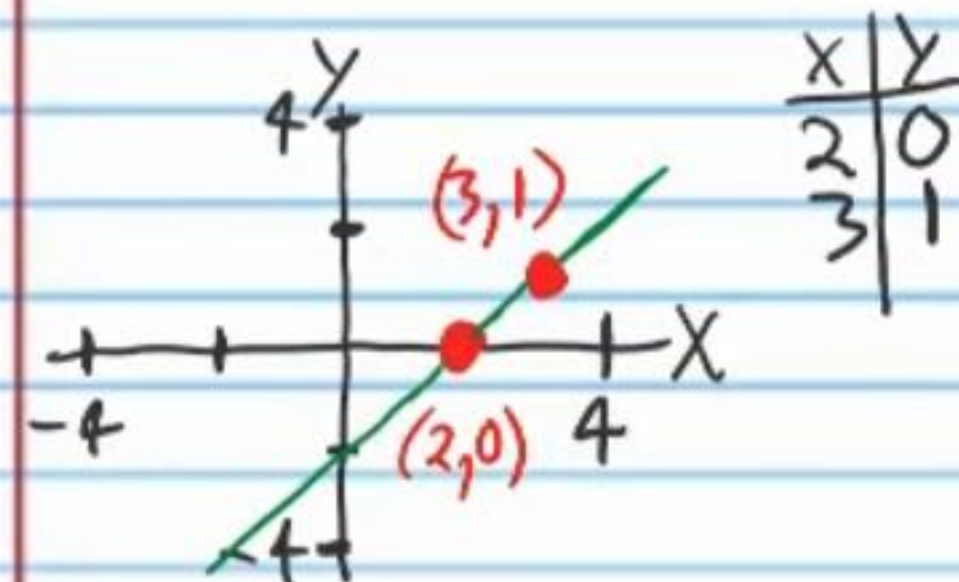
$$\frac{5}{1.4} = \frac{1.4y}{1.4}$$

$$3.57 \approx y$$

$$\boxed{3.6 \approx y}$$

Graph the following equations. Show all work and identify the points you use to graph the equations. Label your graphs properly.

- ⑳ Graph $x = y + 2$.



1st Point

$$x = 0 + 2$$

$$x = 2$$

$$(2, 0)$$

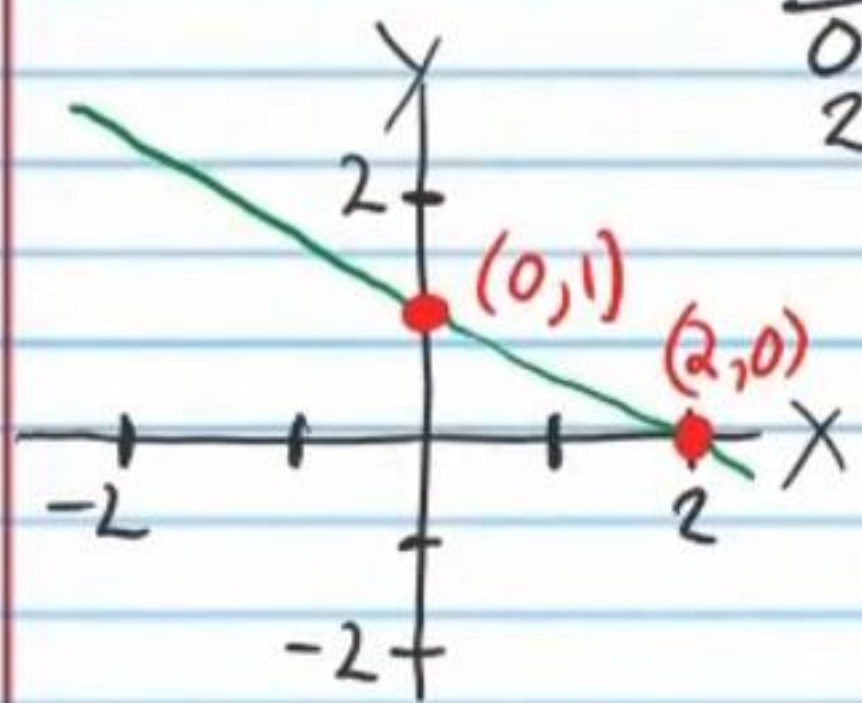
2nd Point

$$x = 1 + 2$$

$$x = 3$$

$$(3, 1)$$

②① $y = 1 - \frac{x}{2}$



X	Y
0	1
2	0

1st Point

$$y = 1 - \frac{0}{2}$$

$$y = 1 - 0$$

$$y = 1$$

(0, 1)

2nd Point

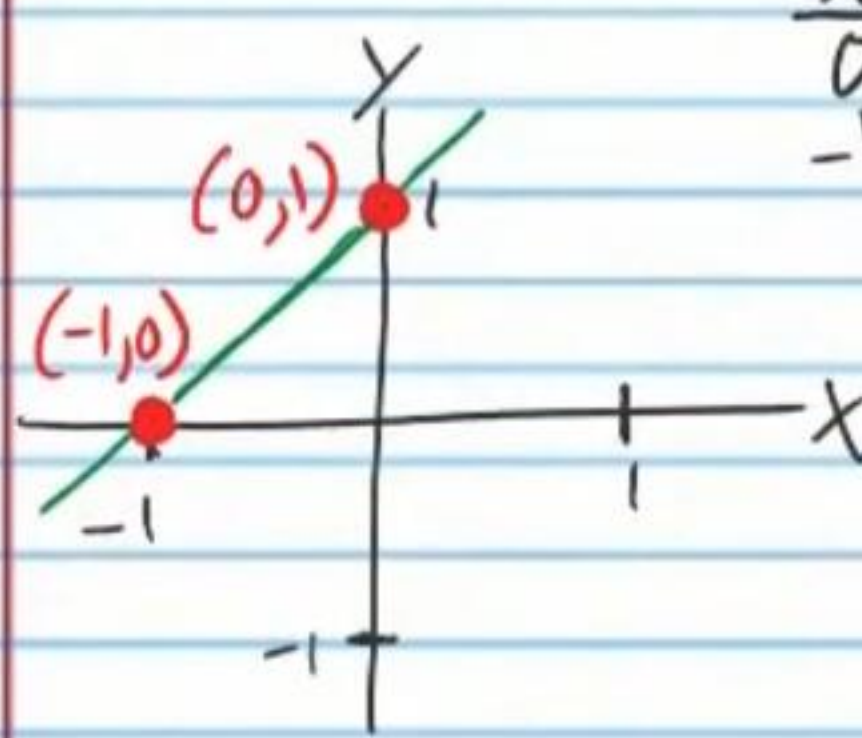
$$y = 1 - \frac{2}{2}$$

$$y = 1 - 1$$

$$y = 0$$

(2, 0)

②② $-x + y = 1$



X	Y
0	1
-1	0

1st Point

$$-0 + y = 1$$

$$0 + y = 1$$

$$y = 1$$

(0, 1)

2nd Point

$$-x + 0 = 1$$

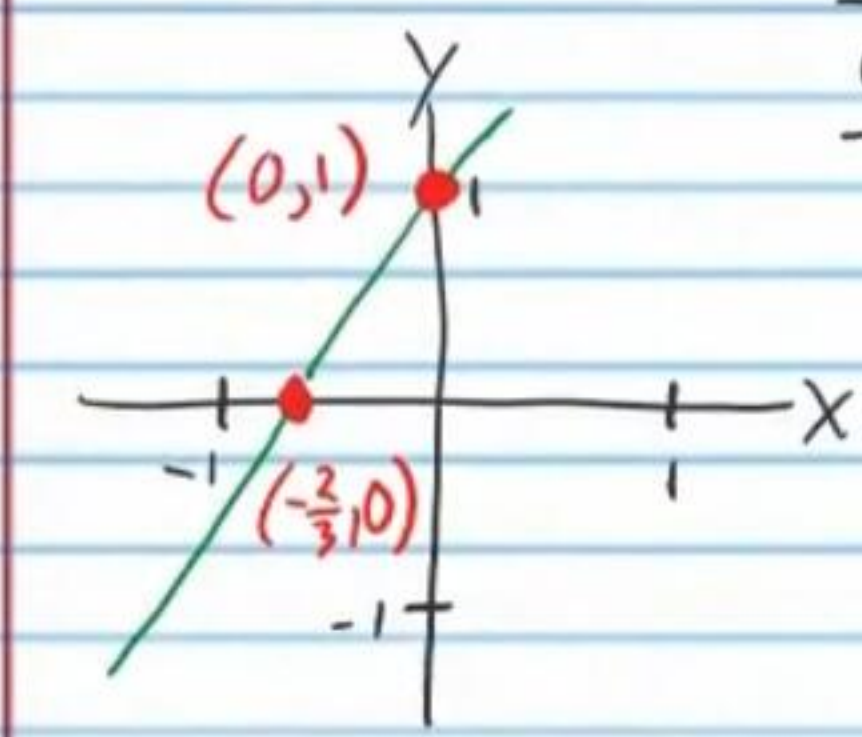
$$-x = 1$$

$$\frac{-1x}{-1} = \frac{1}{-1}$$

$$x = -1$$

(-1, 0)

②③ $-3x + 2y = 2$



X	Y
0	1
-2/3	0

1st Point

$$-3(0) + 2y = 2$$

$$0 + 2y = 2$$

$$\frac{2y}{2} = \frac{2}{2}$$

$$y = 1$$

(0, 1)

2nd Point

$$-3x + 2(0) = 2$$

$$-3x + 0 = 2$$

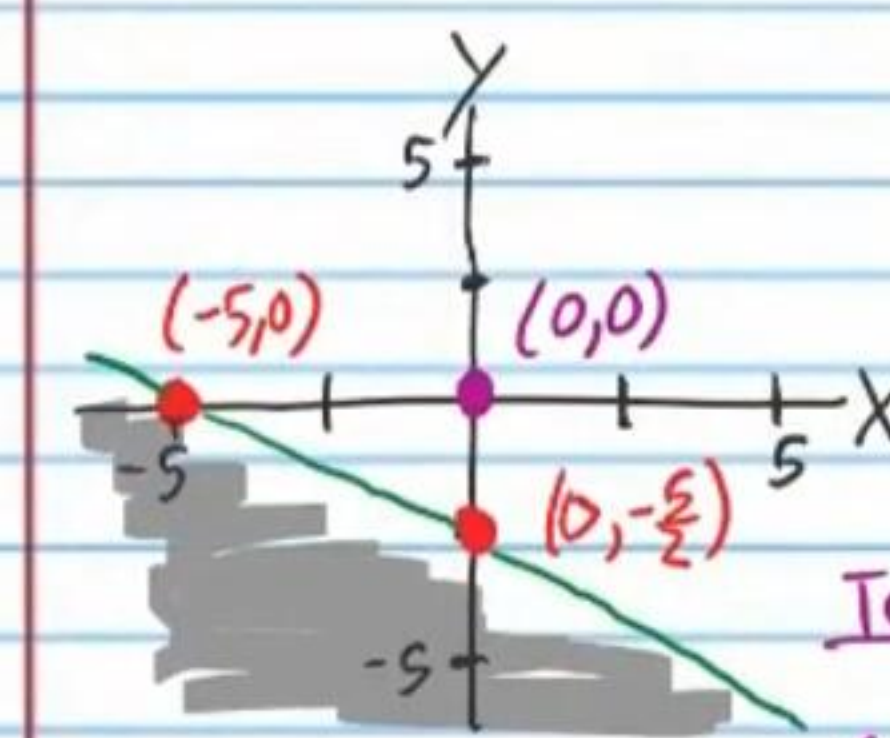
$$\frac{-3x}{-3} = \frac{2}{-3}$$

$$x = -\frac{2}{3}$$

(-2/3, 0)

Graph the following inequalities. Show all work and identify the points you use to graph the inequalities. Label your graphs properly.

②④ $-4 - 2y \geq x + 1$



X	Y
-5	0
0	-5/2

1st Point

$$-4 - 2(0) = x + 1$$

$$-4 - 0 = x + 1$$

$$-1 - 4 = x + 1 - 1$$

$$-5 = x$$

(-5, 0)

2nd Point

$$-4 - 2y = 0 + 1$$

$$-4 - 2y = 1$$

$$4 - 4 - 2y = 1 + 4$$

$$-2y = 5$$

$$\frac{-2y}{-2} = \frac{5}{-2}$$

$$y = -\frac{5}{2}$$

(0, -5/2)

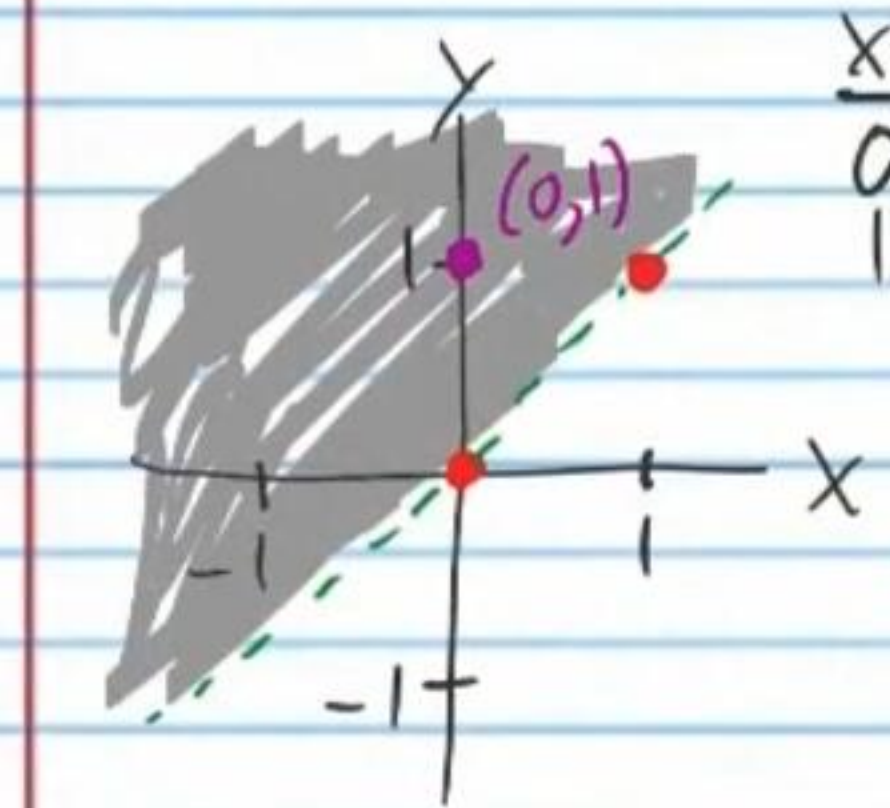
Test Point

$$-4 - 2(0) \geq 0 + 1$$

$$-4 - 0 \geq 1$$

$$-4 \geq 1$$

②⑤ $y > x$



X	Y
0	0
1	1

1st Point

$$y = 0$$

(0, 0)

2nd Point

$$y = 1$$

(1, 1)

Test Point

$$1 > 0$$